

Ex.no:9	MINI PROJECT – SOCIAL MEDIA DATA ANALYTICS

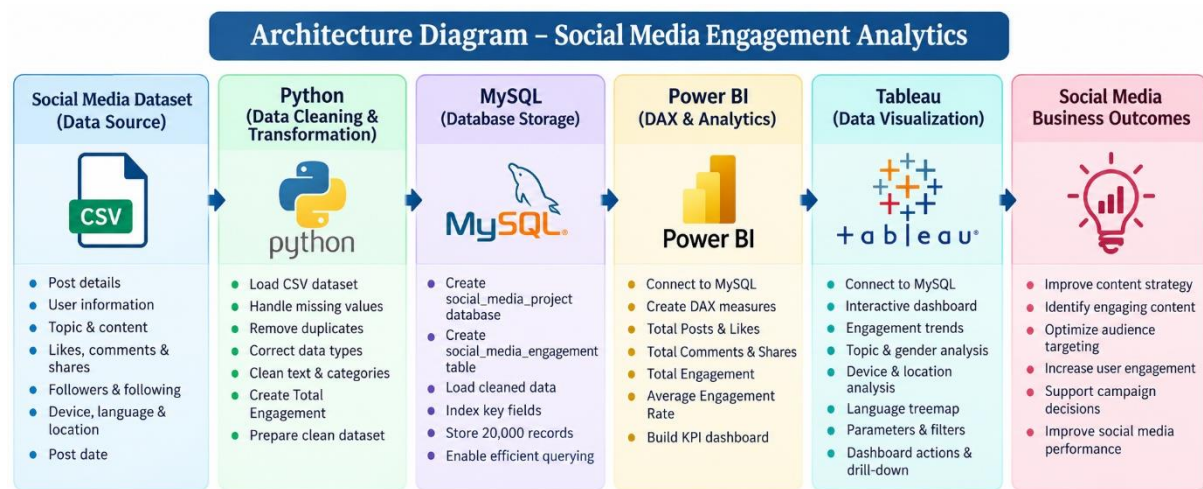
Aim:

To clean, transform, store, and analyze Social Media Engagement data using Python, MySQL, Power BI, and Tableau to derive meaningful insights into post performance, user engagement, user demographics, content topics, devices, locations, languages, engagement trends, and key engagement metrics.

Algorithm:

1. **Load** the Social Media Engagement dataset from the CSV file using **Pandas**.
2. **Clean** the data by handling missing values, removing duplicate records, correcting data types, and eliminating unnecessary attributes.
3. **Transform** the data by standardizing column names and preparing useful fields such as Total Engagement, Engagement Rate, Engagement per Follower, Post Date, User Age, Followers Count, Likes, Comments, and Shares.
4. **Store** the cleaned and transformed dataset in a **MySQL database** named `social_media_project`.
5. **Connect** the MySQL database to **Power BI** and **Tableau** for data analysis and visualization.
6. **Create DAX measures** in Power BI for key metrics such as Total Posts, Total Likes, Total Comments, Total Shares, Total Engagement, and Average Engagement Rate.
7. Develop interactive visualizations to analyze engagement trends, topic-wise engagement, gender-wise engagement, device-wise engagement, location-wise engagement, language-wise engagement, likes, comments, shares, and overall post performance.
8. Implement advanced Tableau features such as calculated fields, parameters, filters, interactive dashboards, dashboard actions, drill-down, and advanced visualizations including line charts, bar charts, pie/donut charts, maps, and treemaps.
9. **Analyze and interpret** the visualizations to identify high-performing topics, user engagement patterns, popular devices, geographic trends, and changes in engagement across different metrics.
10. Generate meaningful business insights from the Power BI and Tableau dashboards to support data-driven understanding of social media performance.
11. **End** the process.

Architecture:



Program:

```
import pandas as pd
```

```
# Load dataset
```

```
df = pd.read_csv("synthetic_social_media_engagement.csv")
```

```
# Create a copy for processing
```

```
social_media = df.copy()
```

```
# Remove duplicate records
```

```
social_media.drop_duplicates(inplace=True)
```

```
# Handle missing values
```

```
social_media["user_name"] = social_media["user_name"].fillna("Unknown")
```

```
social_media["location"] = social_media["location"].fillna("Unknown")
```

```
social_media["topic"] = social_media["topic"].fillna("Unknown")
```

```
social_media["device"] = social_media["device"].fillna("Unknown")
```

```
social_media["language"] = social_media["language"].fillna("Unknown")
```

```
social_media["hashtags"] = social_media["hashtags"].fillna("No Hashtags")
```

```
# Convert date columns
```

```
social_media["account_creation_date"] = pd.to_datetime(  
    social_media["account_creation_date"],  
    errors="coerce"  
)
```

```
social_media["post_date"] = pd.to_datetime(  
    social_media["post_date"],  
    errors="coerce"  
)
```

```
# Convert numerical columns
```

```
numeric_columns = [  
    "user_age",  
    "followers_count",  
    "following_count",  
    "content_length",  
    "likes",  
    "comments",  
    "shares",  
    "engagement_rate"  
]
```

```
for column in numeric_columns:
```

```
    social_media[column] = pd.to_numeric(  
        social_media[column],  
        errors="coerce"  
    )
```

```
# Fill missing numerical values
```

```
for column in numeric_columns:
    social_media[column] = social_media[column].fillna(0)
```

```
# Create derived metric
```

```
social_media["total_engagement"] = (
    social_media["likes"]
    + social_media["comments"]
    + social_media["shares"]
)
```

```
# Create engagement per follower
```

```
social_media["engagement_per_follower"] = (
    social_media["total_engagement"]
    / social_media["followers_count"].replace(0, 1)
)
```

```
# Standardize column names
```

```
social_media.columns = (
    social_media.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)
```

```
# Display dataset information
```

```
print("Original Shape:", df.shape)
print("Cleaned Shape:", social_media.shape)
```

```
print("\nCleaned Dataset:")
```

```
print(social_media.head())
```

```
print("\nMissing Values:")
```

```
print(social_media.isnull().sum())
```

```
print("\nDuplicate Records:")
```

```
print(social_media.duplicated().sum())
```

2. MySQL – Database and Data Loading

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS social_media_project;
```

```
USE social_media_project;
```

Python / Jupyter

```
from sqlalchemy import create_engine
```

```
import pandas as pd
```

```
# MySQL connection details
```

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
    f"mysql+pymysql://{username}:{password}@{host}:{port}/social_media_project"
```

```
)
```

```
# Load cleaned data into MySQL
```

```
social_media.to_sql(
```

```
    "social_media_engagement",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
index=False,  
chunksize=5000
```

```
)
```

```
print("Social Media Engagement data loaded successfully into MySQL.")
```

Verify Data

```
pd.read_sql(  
    "SELECT COUNT(*) AS total_rows FROM social_media_engagement",  
    db_engine  
)
```

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
1 -- Create database  
2 CREATE DATABASE IF NOT EXISTS social_media_project;  
3 USE social_media_project;  
4  
5 -- View tables  
6 SHOW TABLES;  
7  
8 -- Check total records  
9 SELECT COUNT(*) AS total_records  
10 FROM social_media_engagement;  
11  
12 -- View sample data  
13 SELECT * FROM social_media_engagement LIMIT 1000;
```

The 'Result Grid' tab shows the results of the queries. The first query returns 1 row (1 row(s) affected). The second query returns 0 rows (0 row(s) affected). The third query returns 1 row (1 row(s) returned). The fourth query returns 1000 rows (1000 row(s) returned).

post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location	topic	post_content	content_length	hashtags	has_media	post_date	post_id	device	language
P001A709	U1001	@travellertine	Female	24	12500	320	2021-06-14	1	Mumbai	Travel	Exploring the beaches of Goa!	42	#travel #goa	1	2024-05-12	2024-05-12	Mobile	English
P002B14	U1002	@techworld	Male	28	50200	410	2019-11-03	0	Bengaluru	Technology	New smartphone launch amazing!	37	#tech #smartphone	1	2024-05-13	2024-05-13	Desktop	English
P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	1	Delhi	Fashion	Summer collection is finally here!	35	#fashion #style	0	2024-05-13	2024-05-13	Mobile	Hindi
P004D622	U1004	@sportsfan	Male	27	30800	760	2020-09-17	0	Chennai	Sports	What a match! India wins again!	33	#cricket #india	1	2024-05-14	2024-05-14	Mobile	English
P005E828	U1005	@foodieats	Female	25	22100	540	2021-03-10	1	Hyderabad	Food	Tried this amazing biryani today!	37	#food #biryani	1	2024-05-14	2024-05-14	Mobile	Tamil
P006F4R1	U1006	@musiclover	Male	30	63400	890	2018-07-21	1	Kolkata	Music	This song is on repeat!	29	#music #love	0	2024-05-15	2024-05-15	Desktop	English
P007G5T3	U1007	@moviebuff	Female	26	29800	660	2020-12-05	0	Pune	Movies	Just watched an amazing movie!	31	#movies #review	1	2024-05-16	2024-05-16	Mobile	English
P008H7V2	U1008	@lifestylehub	Male	29	41200	520	2019-08-19	1	Chennai	Lifestyle	A productive day!	25	#lifestyle #motivation	0	2024-05-16	2024-05-16	Mobile	English
P009J1X4	U1009	@gamezone	Male	21	17500	310	2022-05-11	0	Mumbai	Gaming	New game release looks awesome!	39	#gaming #new	1	2024-05-17	2024-05-17	Desktop	English
P010K3Y6	U1010	@bookworms	Female	23	26000	470	2021-09-28	1	Bengaluru	Education	Reading this book is a must!	28	#books #reading	0	2024-05-17	2024-05-17	Mobile	English

3. Power BI – DAX Measures

1. Total Posts

Total Posts = `DISTINCTCOUNT(social_media_engagement[post_id])`

2. Total Likes

Total Likes = `SUM(social_media_engagement[likes])`

3. Total Comments

Total Comments = `SUM(social_media_engagement[comments])`

4. Total Shares

Total Shares = `SUM(social_media_engagement[shares])`

5. Total Engagement

Total Engagement =

[Total Likes] + [Total Comments] + [Total Shares]

6. Average Engagement Rate

Average Engagement Rate =

AVERAGE(social_media_engagement[engagement_rate])

7. Average Followers

Average Followers =

AVERAGE(social_media_engagement[followers_count])

8. Average Content Length

Average Content Length =

AVERAGE(social_media_engagement[content_length])

9. Verified Users

Verified Users =

```
CALCULATE(  
    DISTINCTCOUNT(social_media_engagement[user_id]),  
    social_media_engagement[is_verified] = "True"  
)
```

10. Posts with Media

Posts with Media =

```
CALCULATE(  
    DISTINCTCOUNT(social_media_engagement[post_id]),  
    social_media_engagement[has_media] = "True"  
)
```

4. POWER BI Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

Original rows: 20000
Cleaned rows: 20000
Columns: 22

Missing values:

post_id 0
user_id 0
user_name 0
user_gender 0
user_age 0
followers_count 0
following_count 0
account_creation_date 0
is_verified 0
location 0
topic 0
post_content 0
content_length 0
hashtags 0
has_media 0
post_date 0
device 0
language 0
likes 0
comments 0
shares 0
engagement_rate 0
dtype: int64

	post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location	topic	post_content	content_length	hashtags	has_media	post_date	device	language	likes	comments	shares	engagement_rate
0	P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai	Travel	Exploring the beaches of Goa!	42	#travel #goa	True	2024-05-12	Mobile	English	1250	230	120	0.128
1	P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru	Technology	New smartphone launch looks amazing!	37	#tech #smartphone	True	2024-05-13	Desktop	English	3420	560	310	0.085
2	P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi	Fashion	Summer collection is finally here	35	#fashion #style	False	2024-05-13	Mobile	Hindi	980	143	90	0.064
3	P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai	Sports	What a match! India wins again!	33	#cricket #india	True	2024-05-14	Mobile	English	4120	620	450	0.112
4	P005E9Z8	U1005	@foodieats	Female	25	22100	540	2021-03-10	True	Hyderabad	Food	Tried this amazing biryani today!	37	#food #biryani	True	2024-05-14	Mobile	Tamil	2890	410	230	0.096

```
In [10]: pd.read_sql(
         "SELECT COUNT(*) AS total_rows FROM social_media_engagement",
         db_engine
       )
```

	total_rows
0	20000

```
In [11]: pd.read_sql(
         "SELECT * FROM social_media_engagement LIMIT 5",
         db_engine
       )
```

	post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location	topic	post_content	content_length	hashtags	has_media	post_date	device	language	likes	comments	shares	engagement_rate
0	P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai	Travel	Exploring the bea...	42	#travel #goa	True	2024-05-12	Mobile	English	1250	230	120	0.128
1	P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru	Technology	New smartphone L...	37	#tech #smartpho...	True	2024-05-13	Desktop	English	3420	560	310	0.085
2	P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi	Fashion	Summer collection...	35	#fashion #style	False	2024-05-13	Mobile	Hindi	980	143	90	0.064
3	P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai	Sports	What a match! In...	33	#cricket #india	True	2024-05-14	Mobile	English	4120	620	450	0.112
4	P005E9Z8	U1005	@foodieats	Female	25	22100	540	2021-03-10	True	Hyderabad	Food	Tried this amazi...	37	#food #biryani	True	2024-05-14	Mobile	Tamil	2890	410	230	0.096

5 rows × 22 columns



Navigator

Display Options ▾

localhost:3306; social_media_project [1]

☒ ☐ social_media_engagement

social_media_engagement



post_id	user_id	user_name	user_gender	user_age	followers_count	following_count	account_creation_date	is_verified	location
P001A7K9	U1001	@travelwithme	Female	24	12500	320	2021-06-14	True	Mumbai
P002B3L4	U1002	@techworld	Male	28	50200	410	2019-11-03	False	Bengaluru
P003C8M1	U1003	@fashionista	Female	22	18700	980	2022-01-25	True	Delhi
P004D6Q2	U1004	@sportsfan	Male	27	30800	760	2020-09-17	False	Chennai
P005E9Z8	U1005	@foodieats	Female	25	22100	540	2021-03-10	True	Hyderabad
P006F4R1	U1006	@musiclover	Male	30	63400	890	2018-07-21	True	Kolkata
P007G5T3	U1007	@moviebuff	Female	26	29800	660	2020-12-05	False	Pune
P008H7V2	U1008	@lifestylehub	Male	29	41200	520	2019-08-19	True	Chennai
P009J1X4	U1009	@gamerzone	Male	21	17500	310	2022-05-11	False	Mumbai
P010K3Y6	U1010	@bookworm	Female	23	26000	470	2021-09-28	True	Bengaluru

The preview has been truncated due to size limits.

Select Related Tables

Load

Transform Data

Cancel

FileHomeHelpTable tools

NameSocialMediaData

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Share

Auto recovery contains some recovered files that haven't been opened.

View recovered files

Data

post_id

user_id

user_name

user_gender

user_age

followers_count

following_count

f18f3713-819d-4092-9ffc-d22ce025ea90

48be9876-b686-4b07-9998-cf0e5cfb5976

johnny54

Other

47

73361

36

3c08b703-dbab-4720-8efb-ca7f5bae978d

fd480ed6-95f9-4a77-a12c-2a8f369cb741

paigetaylor

Other

59

37225

97

303ceca3-e53f-4e7f-a677-ac9187daeecc

f4d6a67a-4f72-4974-9b31-2c097c8a3608

brianromero

Other

25

97410

45

be9b6245-3224-4e70-a87b-6b596c8a2b0e

5885c6da-8db7-4ee3-ad70-4d692095f0d2

michael82

Other

23

20156

85

1df59ef2-e8a1-4415-9fad-a23e09f48a9f

3952b33e-1181-4251-959d-4cdeec17e32

youngcharles

Other

17

56004

89

fedb8ce2-3daa-4c76-bff3-ddabb692dcd3

8fa34d26-2ec1-4de3-9e11-0040fb2d0ec9

lynnjames

Other

41

83648

10

75ee7ea5-f00b-44c9-b290-3162d4a04e72

4f750d75-c109-408b-9f6e-2c0c8b6adbac

laronkevin

Other

27

14628

42

bed9d266-6d58-41f7-9d49-1e84bdf4198a

1ae5cada-bdf9-4051-8f33-a826b9ba1342

robertpatel

Other

18

38163

19

03ffc686-3574-4d7e-8b8b-54321717506d

c2d57ddc-315b-4f08-b9b8-db4b14f526d6

laurakaufman

Other

46

24828

76

cfb472d4-186f-4984-abdc-a60aa3d6d2d9

142649ca-b5ed-415d-821e-6904b11b994c

jacquelinemartin

Other

26

7467

11

c26167ef-7ccd-49b4-b790-061c41e6ddcc

50880738-5e2f-4e1e-9b80-7146010fb16f

tjimenez

Other

38

17531

19

7a88d278-dfbc-4fd7-8633-72173d38af63

57e5afdf-44e0-4cda-904a-8cda79ede7e6

hannahbell

Other

36

27276

76

d19211e1-e692-438b-b6c9-117bc39aa5ef

f4a44c11-51fb-4766-8952-300f5467a672

jennifersmith

Other

34

88659

59

09cc638d-dab6-4100-8501-4796fa7b5ba9

5885c6da-8db7-4ee3-ad70-4d692095f0d2

michael82

Other

23

20156

85

8956225c-918a-4533-ad12-ec3d0b0d23d8

1310de0c-2ce2-4a65-a457-031fa1f73391

collinsnatalie

Other

24

10938

30

11fb94825-2195-4804-8367-6e6993036abd

aac928d0-ab46-43e4-9e04-2689904f033

crystal53

Other

31

26548

70

c571d511-637b-4a7f-ad74-2eab1139a0a5

4113180c-30d5-447e-a028-e05f56fdd2c5

khammond

Other

37

1946

46

6b56722e-3493-46b4-9e35-26dcb439d9b

cebc179a-6c8e-4e36-b2d5-a2bef35b9f03

bestchristopher

Other

53

65594

44

2c4e3892-324c-4cbf-bf33-1fe45f082746

6597eb99-6532-4bf1-8f11-a4ff85540d6a

smithpaul

Other

35

48156

71

f3c0e837-6ab7-4b03-8682-2117151f8f74

90711904-6c35-4361-917b-61d1eea8caa1

smason

Other

36

82450

86

08f712ec-5332-430f-bc71-0a24d798fb77

3f9c7a5f-6b6f-4240-9a50-6a3383cd331

wcochrane

Other

16

27875

51

5c9e73be-705f-4ce2-9965-a9d65eb57b26

ba50d987-70b7-46b1-9c06-6955914229c7

javier42

Other

58

16376

47

84b42665-b326-4f50-8a87-c0adebde0c11

3374fa25-c8bf-492e-9706-531b7719e1ad

ohall

Other

48

4398

97

a2e6e0d-0847-4f07-ab67-3b8abb5b7063

ae42ede5-fce0-454a-8c89-68ed6c87cdcb

jacobsonsamantha

Other

39

12334

41

597d8cd8-53b7-42f5-a728-90888a109a6f

2b77190a-3f33-4e5e-a137-bbd04494b3fb

susanmurohv

Other

54

13072

26

Table: SocialMediaData (20,000 rows)

Data Visualization Using Power BI

The Social Media Engagement dataset was imported into Power BI Desktop to analyze post performance, user engagement, demographics, devices, locations, languages, and engagement trends.

Interactive KPI cards and charts were created to identify popular topics, user preferences, and overall social media performance.

Visualization 1 — Engagement by Topic Chart: Bar Chart

Shows the total engagement for different social media topics.

This visualization helps identify the most engaging content categories, such as Technology, Travel, Sports, Food, and Fashion.

Visualization 2 — Engagement by Location Chart: Map

Shows the distribution of social media engagement across different cities in India.

This visualization helps identify locations with the highest user engagement.

Visualization 3 — Gender-wise Engagement Chart: Bar Chart

Shows the comparison of engagement between different user genders:

- Male
- Female

This helps analyze whether engagement patterns differ based on gender.

Visualization 4 — Device-wise Engagement Chart: Pie Chart

Shows the distribution of social media engagement based on the devices used:

- Mobile
- Desktop
- Tablet

This visualization helps identify the most commonly used devices for social media activity.

Visualization 5 — Language-wise Engagement Chart: Treemap

Shows engagement across different languages used in social media posts.

This helps identify popular languages and understand audience preferences.

Visualization 6 — Engagement Trend Chart: Line Chart

Shows how social media engagement changes over time.

This visualization helps identify increasing or decreasing engagement patterns across different dates or months.

Visualization 7 — Total Engagement KPI Card: Card

Displays the total engagement generated through likes, comments, and shares.

This provides an overall measure of social media performance.

Visualization 8 — Average Engagement Rate KPI Card: Card

Shows the average engagement rate of all posts.
This helps evaluate the overall effectiveness of user interaction with the content.

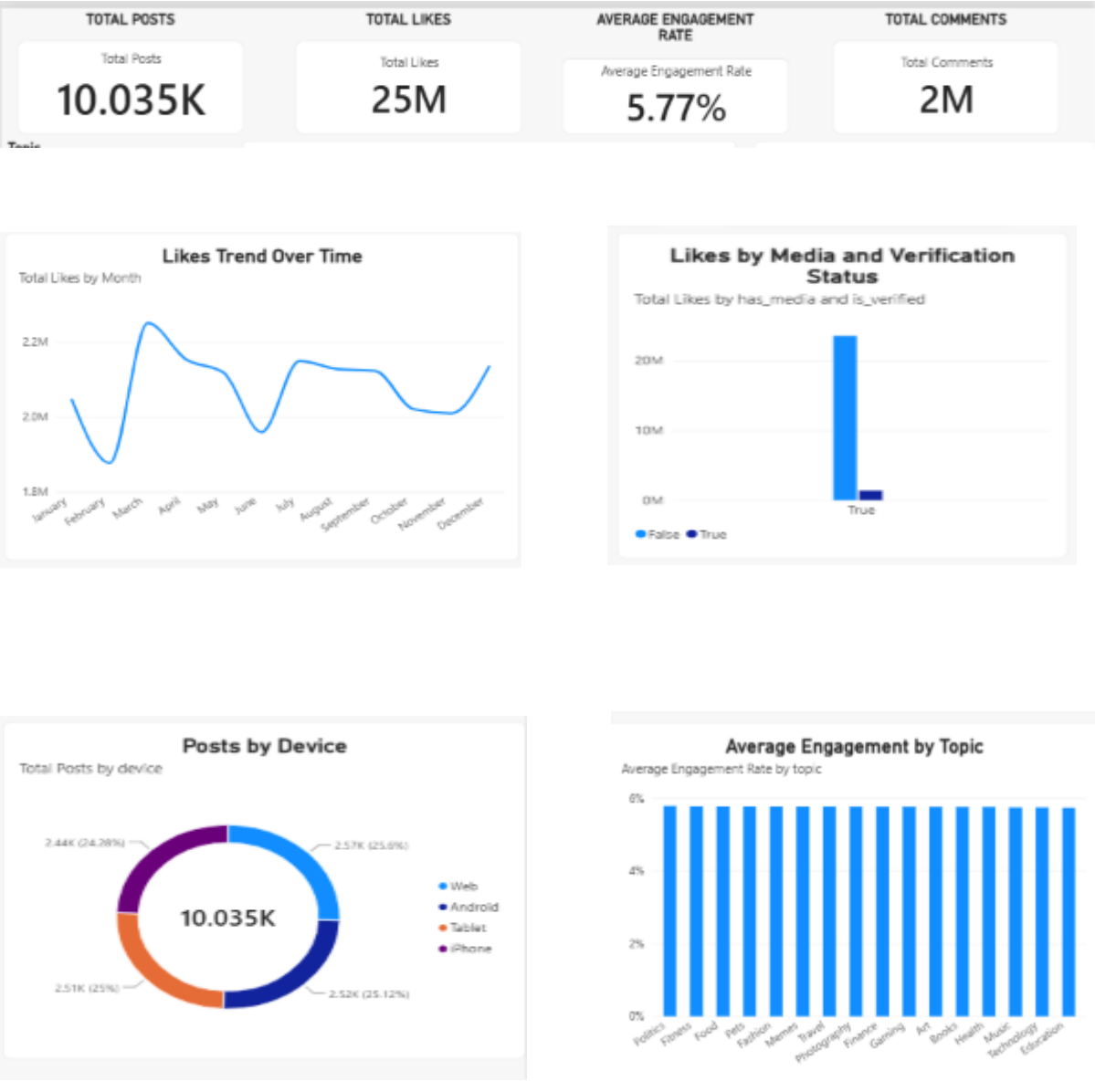
Visualization 9 — Total Posts KPI Card: Card

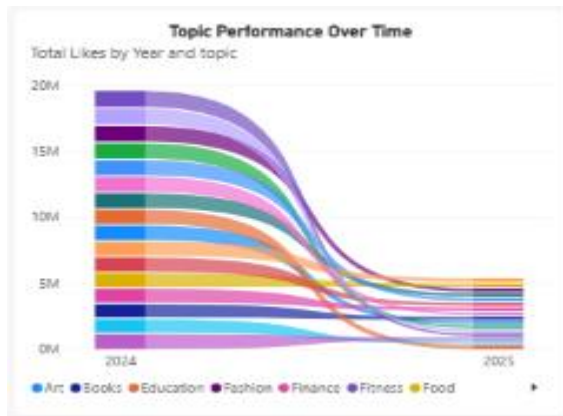
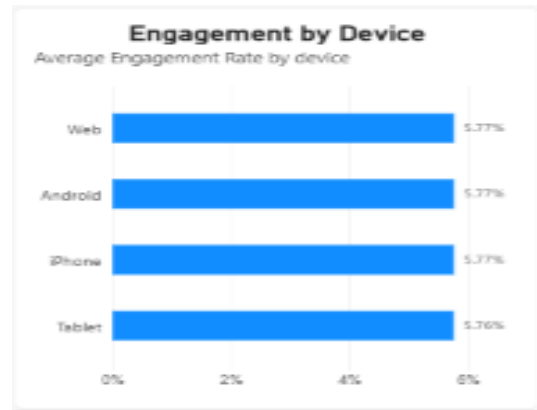
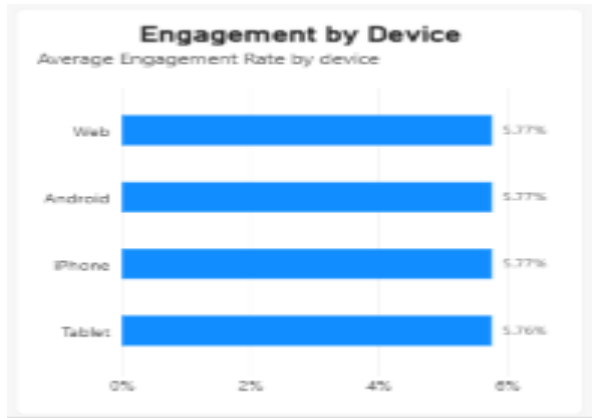
Displays the total number of social media posts in the dataset.
This provides an overview of the dataset size and posting activity.

Visualization 10 — Total Likes, Comments and Shares KPI Cards: Cards

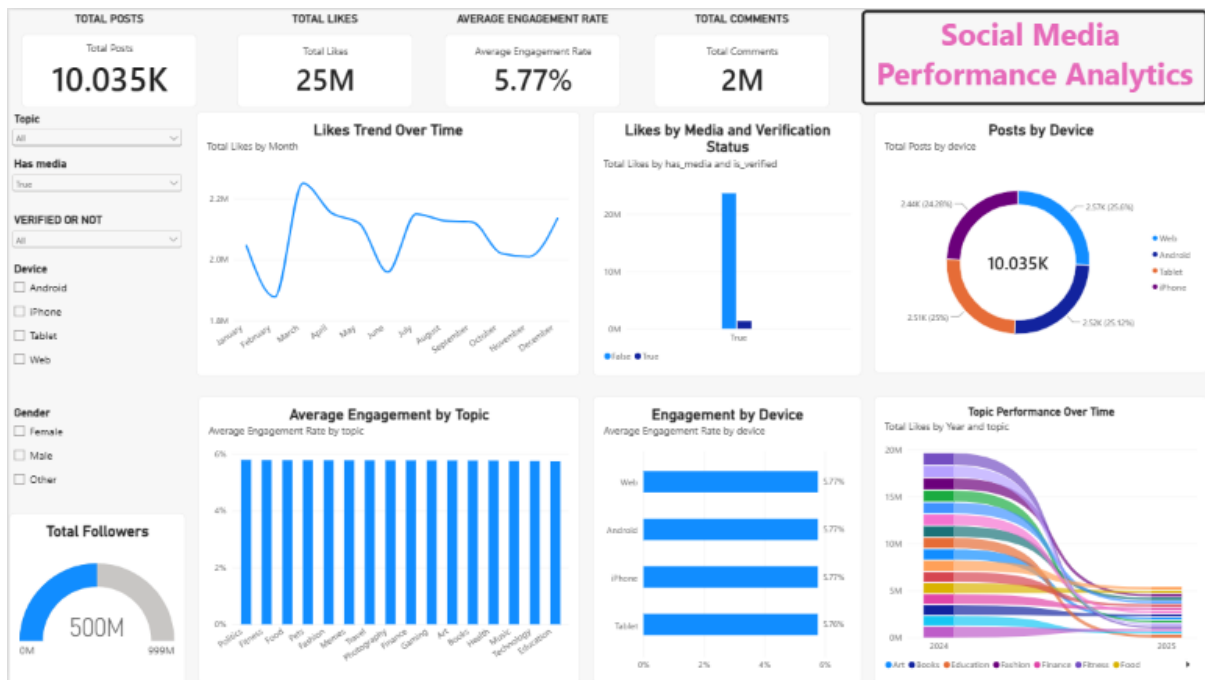
Shows the total number of likes, comments, and shares received by all posts.

OUTPUT :





SOCIAL MEDIA ANALYTICS POWER BI DASHBOARD



5. Tableau Dashboard

Created visuals in Tableau in separate sheets and created a dashboard

File Data Server Window Help

social_media_engagement

Connection: ☒ Live ☐ Extract Filters: 0 | Add

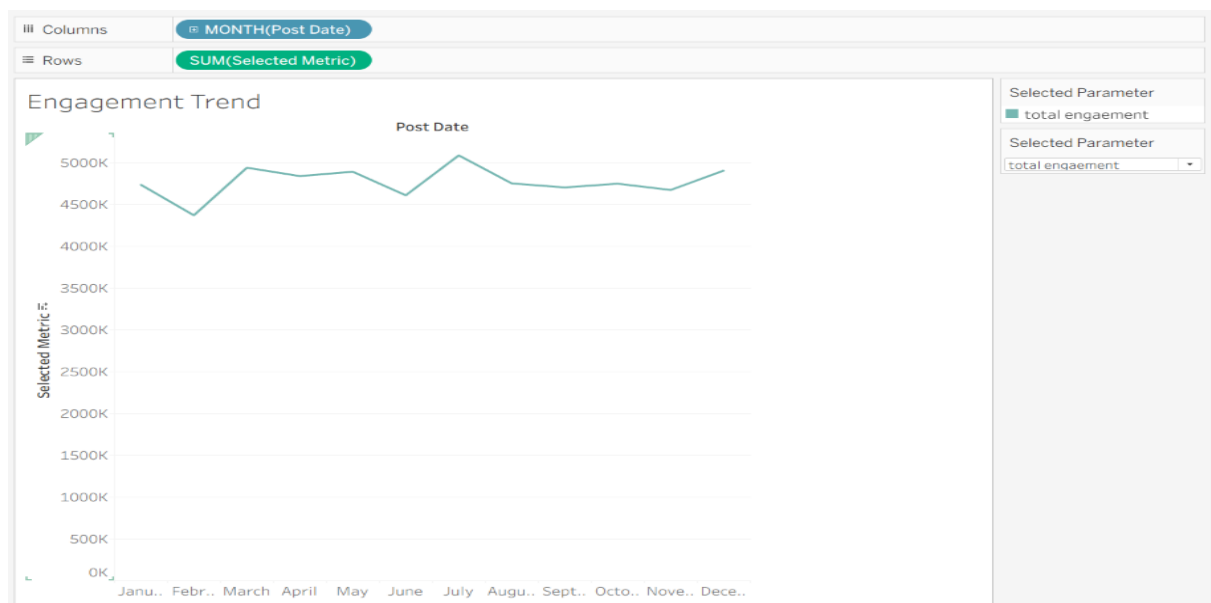
Connections: social_media_engagement MySQL

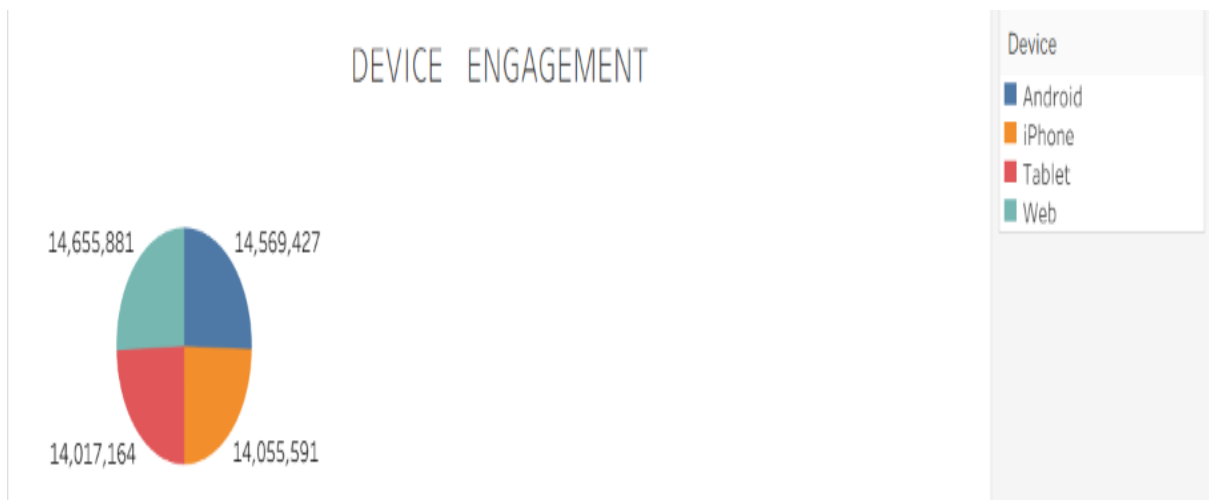
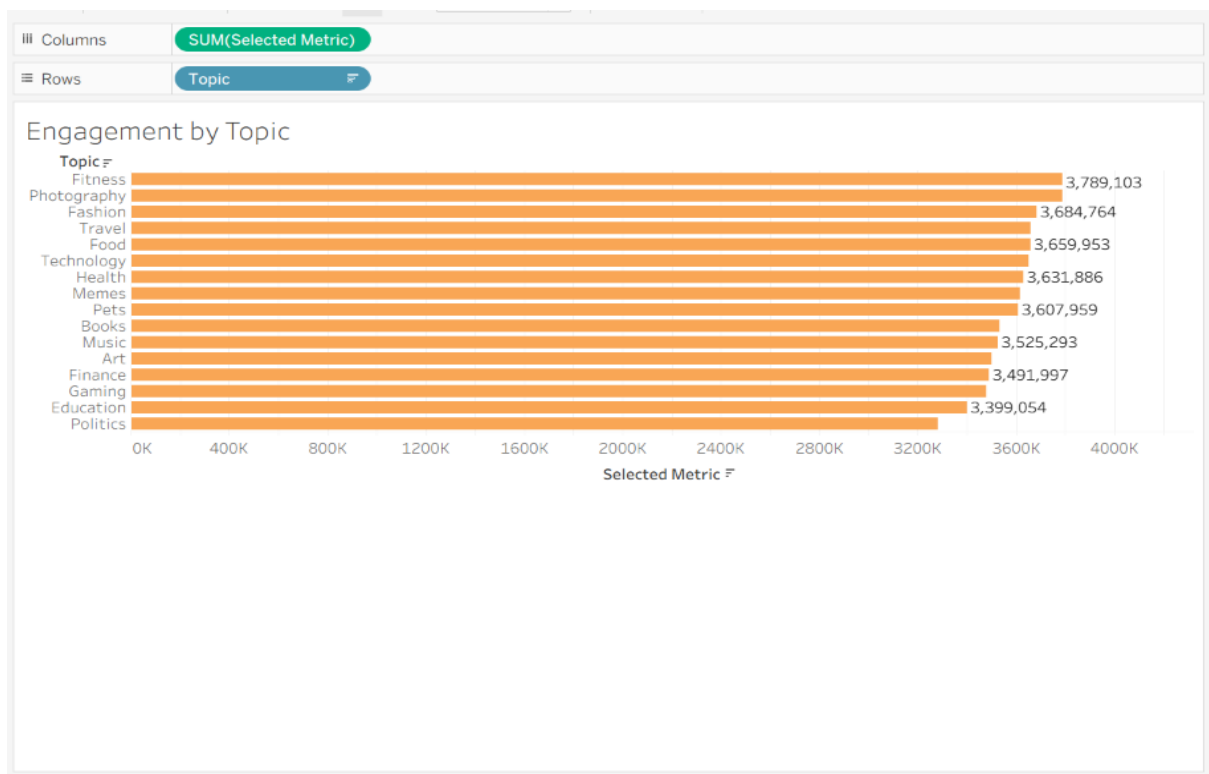
Database: social_media_project

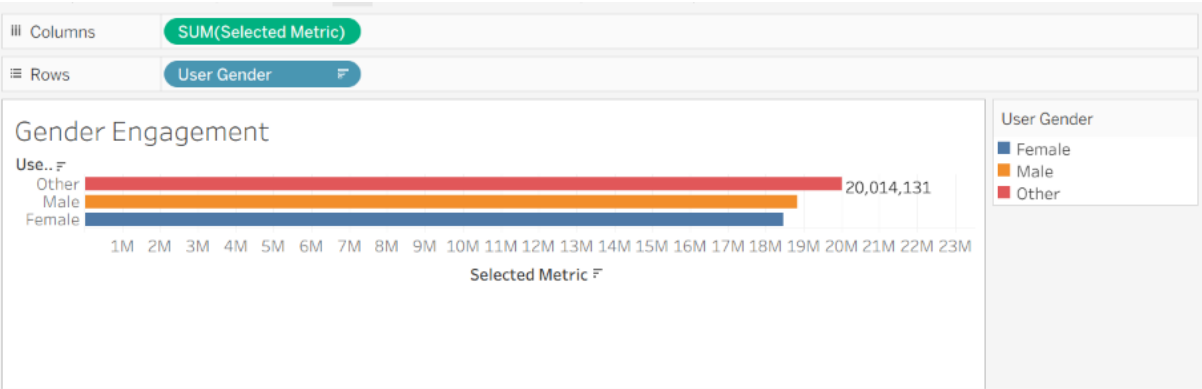
Table: social_media_engagement

social_media_engagement 22 fields 20,000 rows 100 rows

Name	social_media_engagement	post_id	user_id	user_name	user_gender	user_age	followers_count	account_creation_date	is_verified	location	topic
Description	No description available.	P001A7K9	U1001	@travelwithme	Female	24	12500	2021-06-14	True	Mumbai	Travel
Fields		P002B3L4	U1002	@techworld	Male	28	50200	2019-11-03	False	Bengaluru	Technology
		P003C8M1	U1003	@fashionista	Female	22	18700	2022-01-25	True	Delhi	Fashion
		P004D6Q2	U1004	@sportsfan	Male	27	30800	2020-09-17	False	Chennai	Sports
		P005E9Z8	U1005	@foodieeats	Female	25	22100	2021-03-10	True	Hyderabad	Food
		P006F4R1	U1006	@musiclover	Male	30	63400	2018-07-21	True	Kolkata	Music
		P007G5T3	U1007	@moviebuff	Female	26	29800	2020-12-05	False	Pune	Movies
		P008H7V2	U1008	@lifestylehub	Male	29	41200	2019-08-19	True	Chennai	Lifestyle
		P009J1X4	U1009	@gamerzone	Male	21	17500	2022-05-11	False	Mumbai	Gaming
		P010K3Y6	U1010	@bookworm	Female	23	26000	2021-09-28	True	Bengaluru	Education

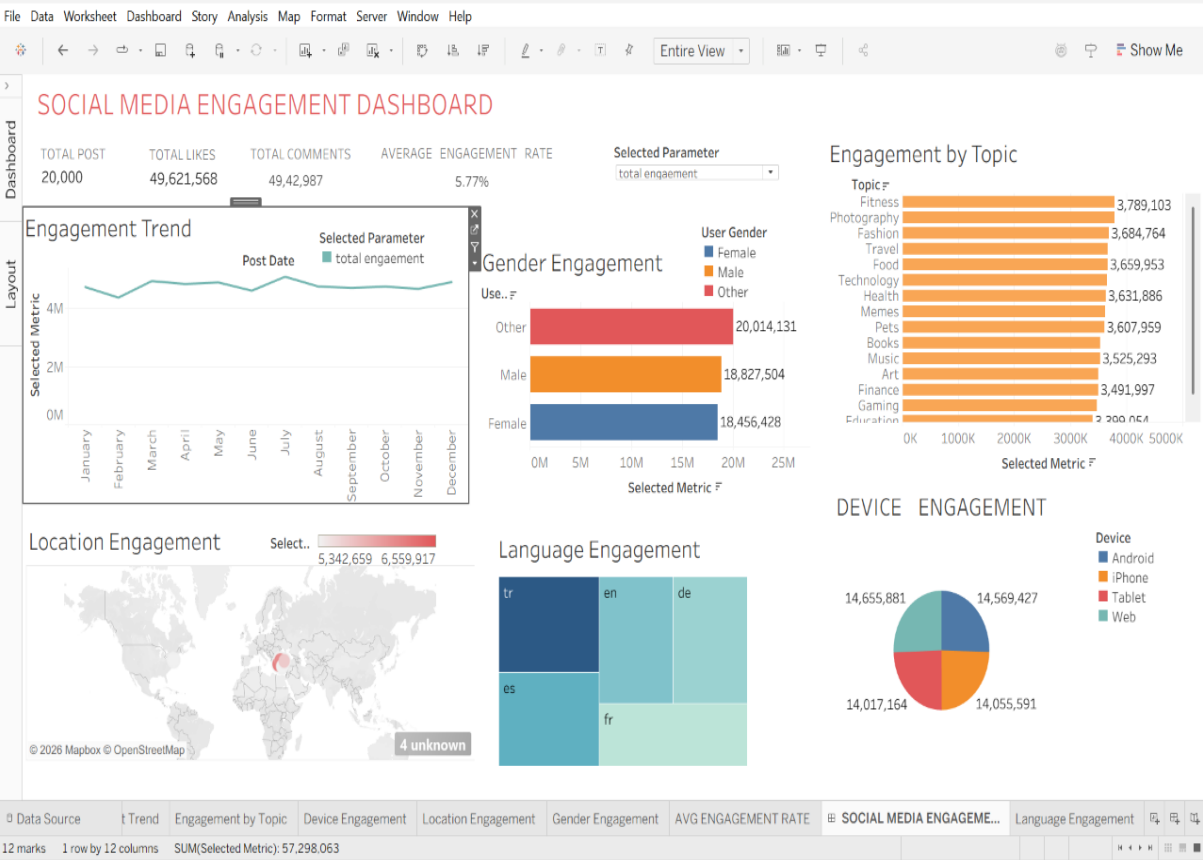






OUTPUT :

SOCIAL MEDIA ANALYTICS TABLEAU DASHBOARD



Result: